Solar Radio Burst Research in Taiwan

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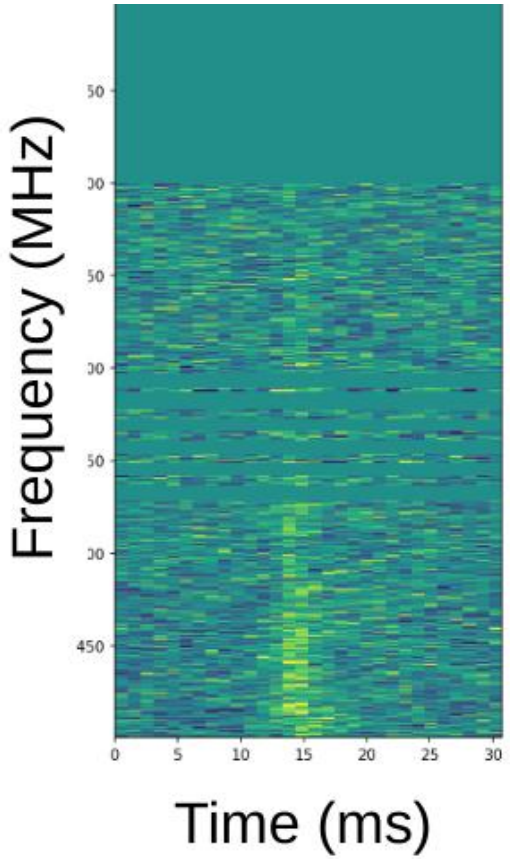
2026.04.11

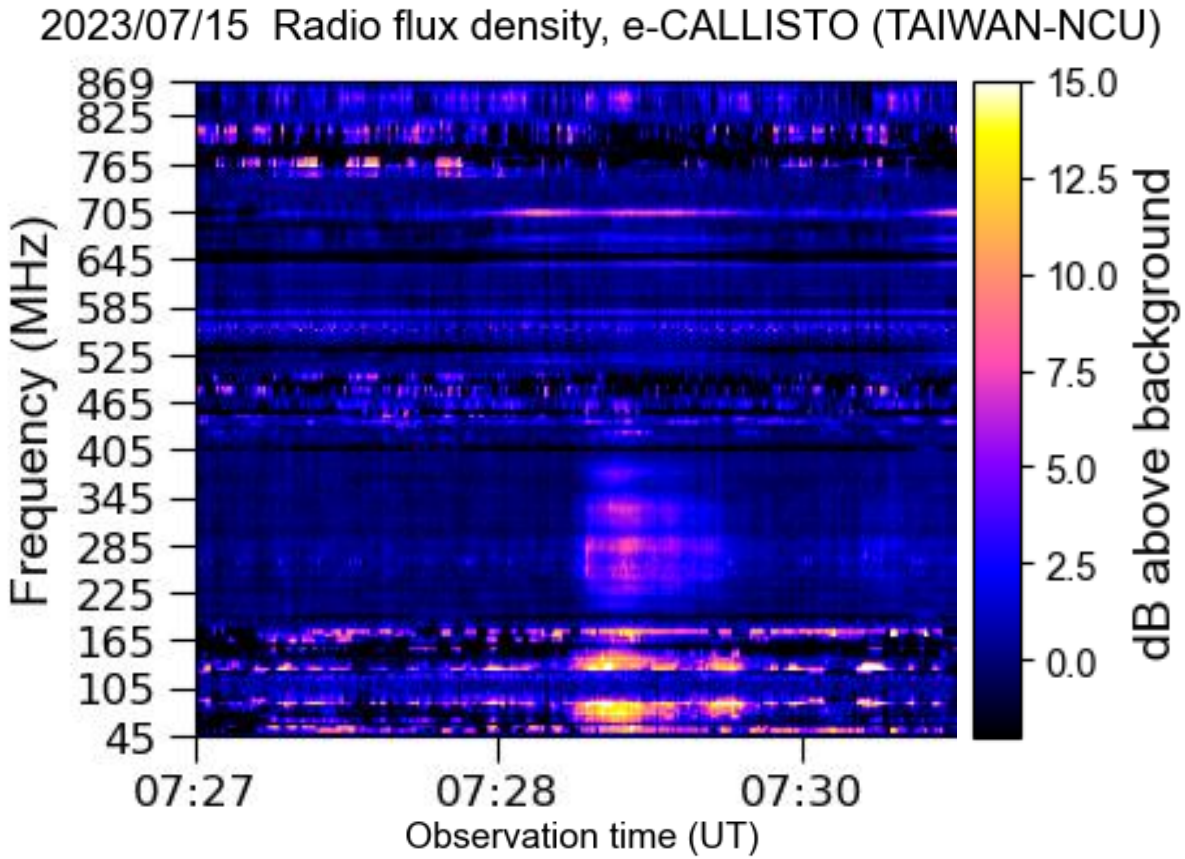


Fast Radio Burst vs. Solar Radio Burst

Both are short-lived, intense pulses of electromagnetic radiation in the radio frequency spectrum.

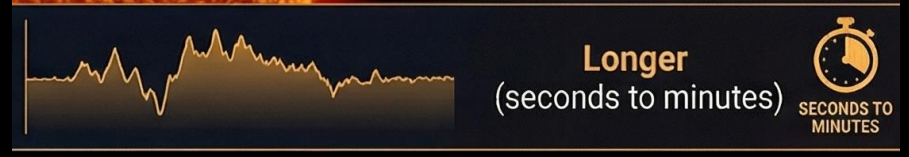
What are the differences and similarities between them?





https://lesia.obspm.fr/cesra/highlights/highlight07-1.html

Differences and Similarities between FRB and SRB

	Fast Radio Burst (FRB)	Solar Radio Burst (SRB)
Origin		
Duration		
Energy	**Extreme** 10^{42} erg/s (≅3 days of total solar output)	**High (Local)** 10^{15} erg
Classification	**Repeaters vs. Non-repeaters** Based on whether multiple bursts are observed from the source.	**Type I to Type V** Based on frequency drift characteristics in dynamic spectra.
Mechanism	**Coherent** Plausible sources include magnetars, neutron star mergers, or synchrotron masers.	**Coherent & Incoherent** Mostly plasma emission or electron-cyclotron maser emission.
Observation	**Ground-base** **(**Space satellites monitor high-energy counterparts.)	**Ground-base & Spaceborne** (Both of them monitor radio and X-ray counterparts.)
Scientific Application	**Cosmic Probes** Used to study dark matter, dark energy, and the intergalactic medium.	**Space Weather** Used for forecasting solar storms and diagnostics of the solar atmosphere.

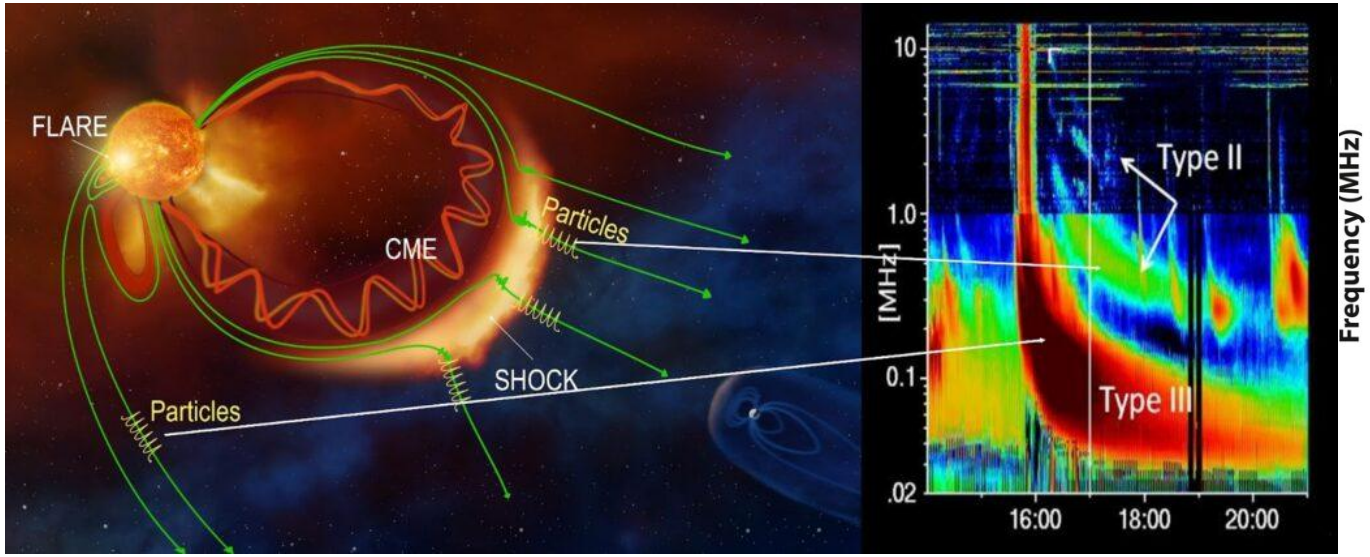
Outline

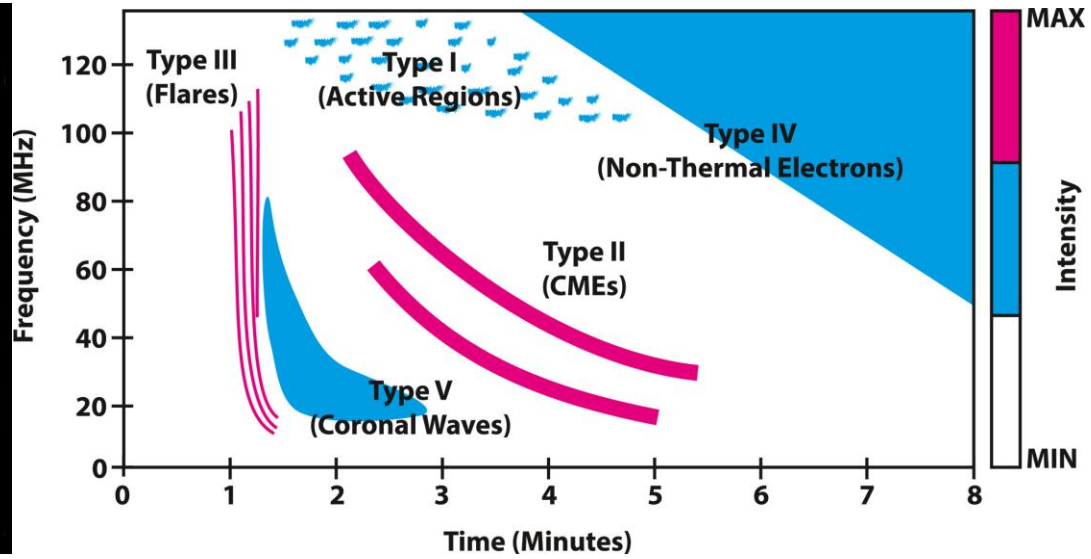
- Differences and Similarities between FRB and SRB
- Introduction of Solar Radio Burst (SRB)
- Current status of SRB research in Taiwan
  - Ground-base
    - e-CALLISTO
    - BURSTT
  - Spaceborne
    - Parker Solar Probe

https://www.dias.ie/cosmicphysics/astrophysics/astro-surround/
Akhavan-Tafti et al. (2025)

Solar Radio burst (SRB)

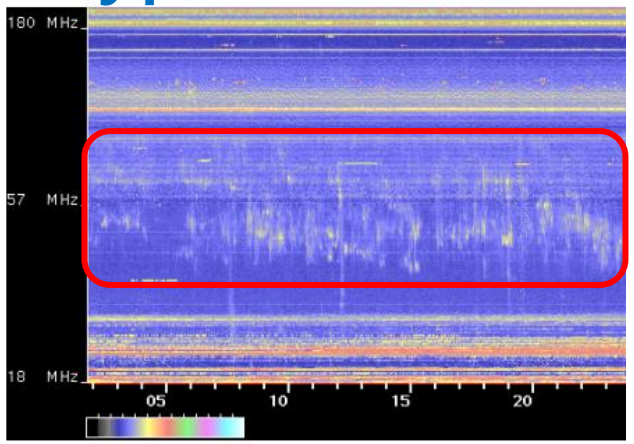
- An intense, transient emission of radio waves from the Sun, caused by energetic electron beams and often associated with solar flares and coronal mass ejections (CMEs).
- Usually could classified into five types based on the signature shown on the dynamic spectrum.



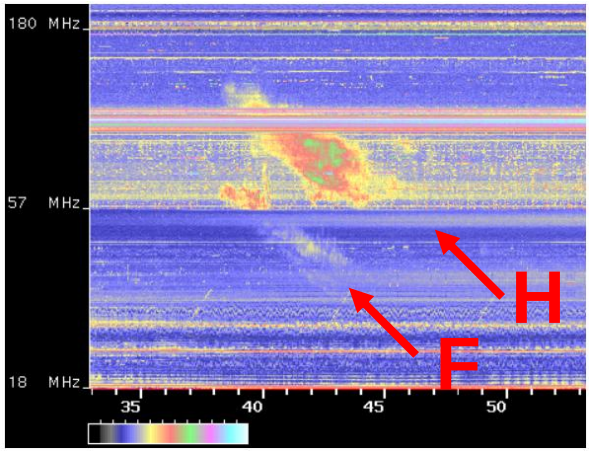


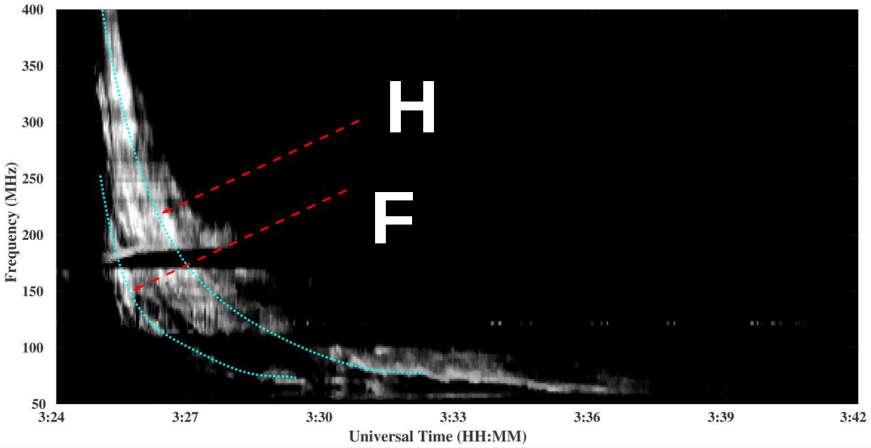
Examples for SRB Classification

■Type I

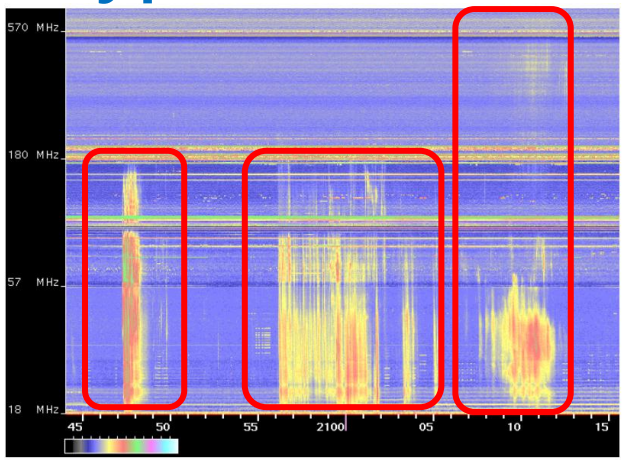


■Type II (Fundamental & Harmonic)

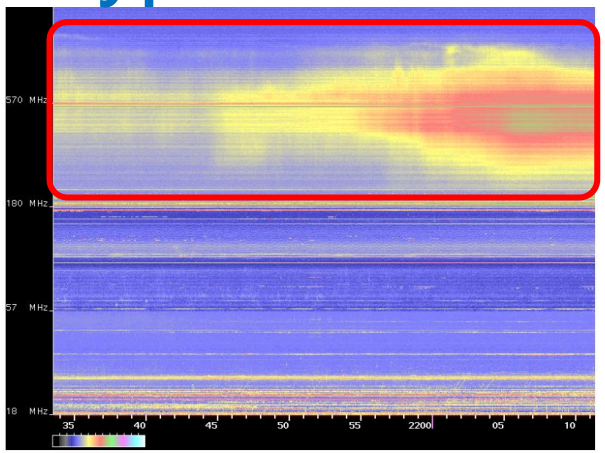




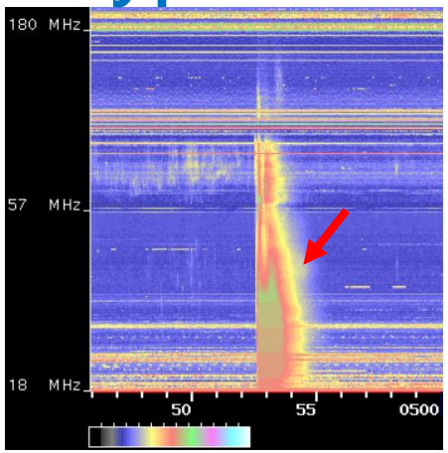
■Type III



■Type IV



■Type V



SRB Classification

TYPE	CHARACTERISTICS	DURATION	FREQUENCY RANGE	ASSOCIATED PHENOMENA
I	Short, narrow-bandwidth bursts. Usually occur in large numbers with underlying continuum.	Single burst: ~ 1 second Storm: hours - days	80 – 200 MHz	Active regions, flares, eruptive prominences.
II	Slow frequency drift bursts. Usually accompanied by a (usually stronger intensity) second harmonic.	3- 30 minutes	Fundamental: 20 – 150 MHz	Flares, proton emission, magnetohydrodynamic shockwaves.
III	Fast frequency drift bursts. Can occur singularly, in groups, or storms (often with underlying continuum). Can be accompanied by a second harmonic	Single burst: 1 - 3 seconds Group: 1 -5 minutes Storm: minutes - hours	10 kHz – 1 GHz	Active regions, flares.
IV	Stationary Type IV: Broadband continuum with fine structure	Hours - days	20 MHz – 2 GHz	Flares, proton emission.
	Moving Type IV: Broadband, slow frequency drift, smooth continuum.	30 – 2 hours	20 – 400 MHz	Eruptive prominences, magnetohydrodynamic shockwaves.
	Flare Continua: Broadband, smooth continuum.	3 – 45 minutes	25 – 200 MHz	Flares, proton emission.
V	Smooth, short-lived continuum. Follows some type III bursts. Never occur in isolation.	1-3 minutes	10 - 200 MHz	Same as type III bursts.

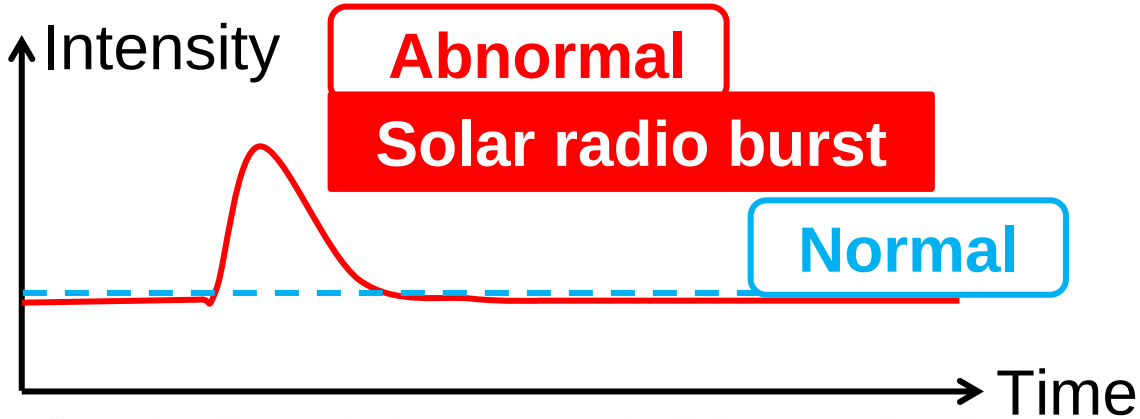
Covered in BURSTT observed frequency windows

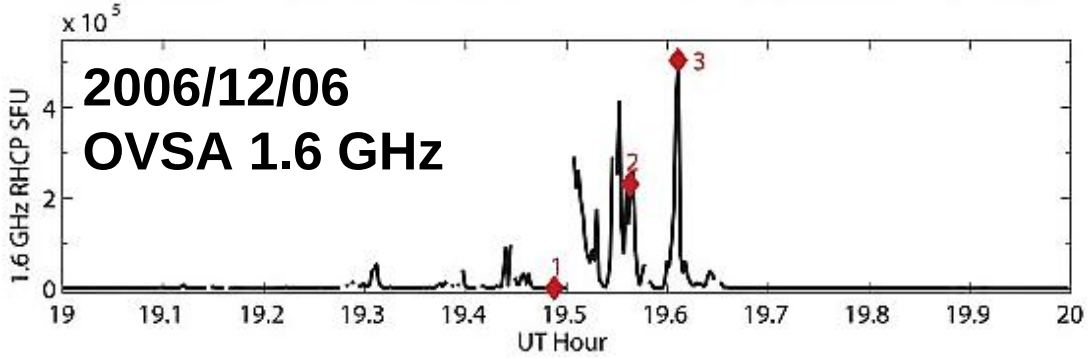
NOTES: In nearly all cases, drifting bursts drift from high to low frequencies.
The Frequency Range is the typical range in which the bursts appear – not their bandwidth.
The sub-types of type IV are not universally agreed upon and are thus open to debate.

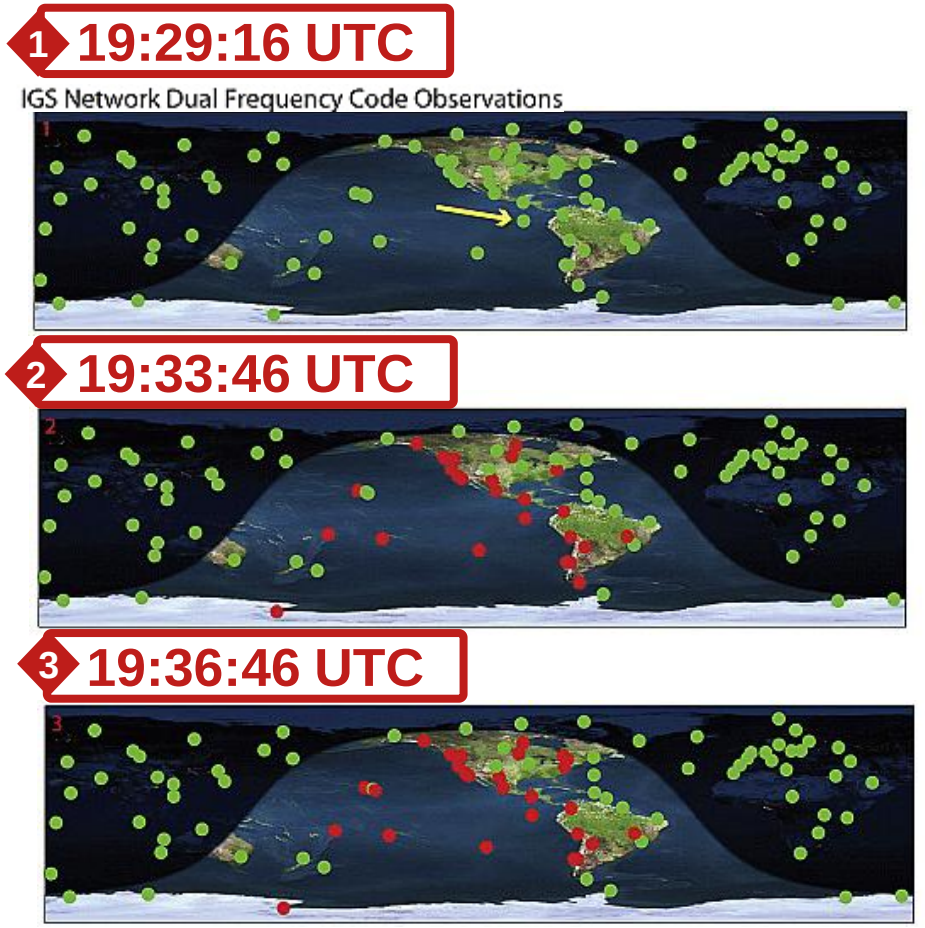
Cerruti et al. (2008)
Kintner Jr. et al. (2009)

SRB as A Disturbance of GNSS Receivers

- During strong solar radio bursts, some GNSS ground receivers don't provide a great positioning solution for user.



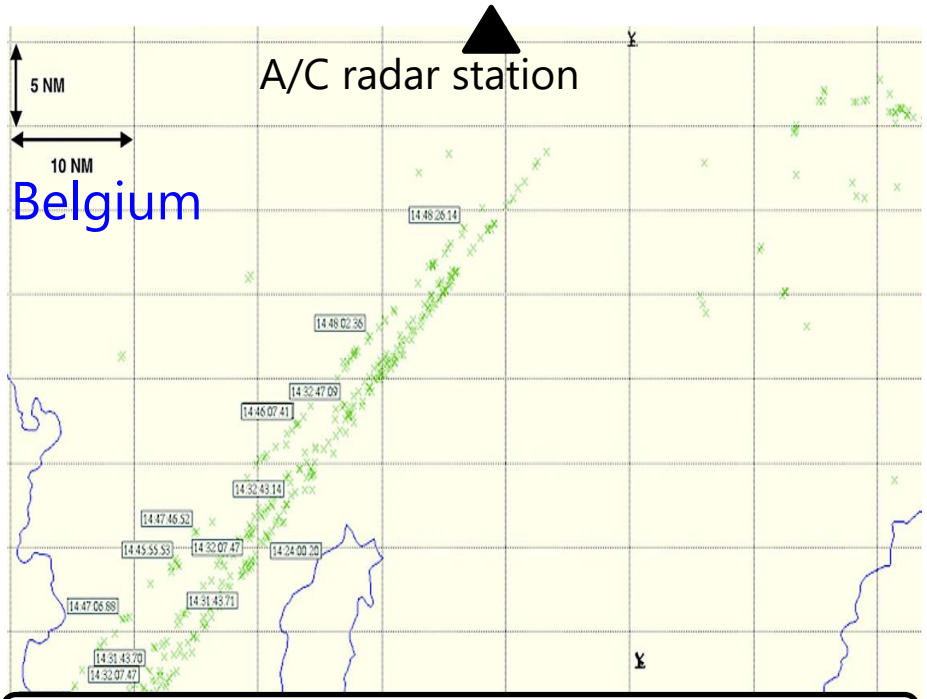




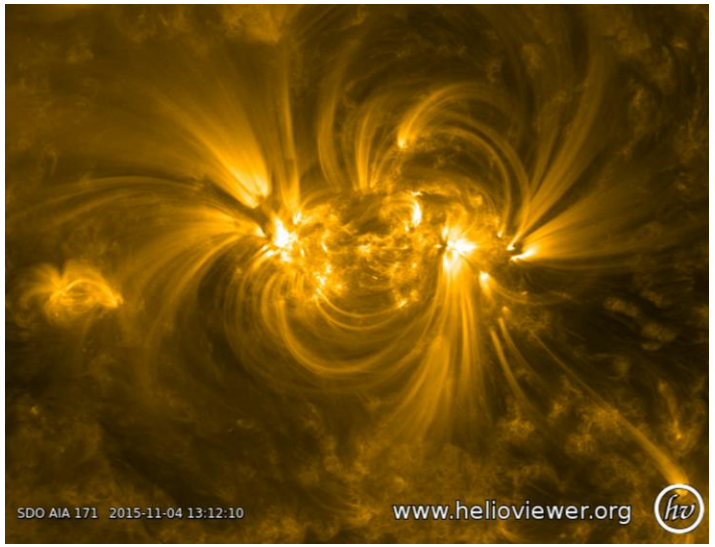
Marqué et al. (2018)

SRB as A Disturbance of Aeronautical Radionavigation

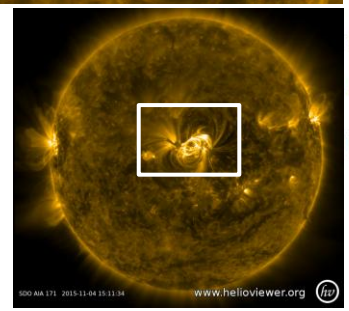
- Air traffic control (ATC) systems located on the southern part of Sweden could not display proper information to the air traffic controllers because of a ghost echoes in straight lines in the direction of the Sun and took place.

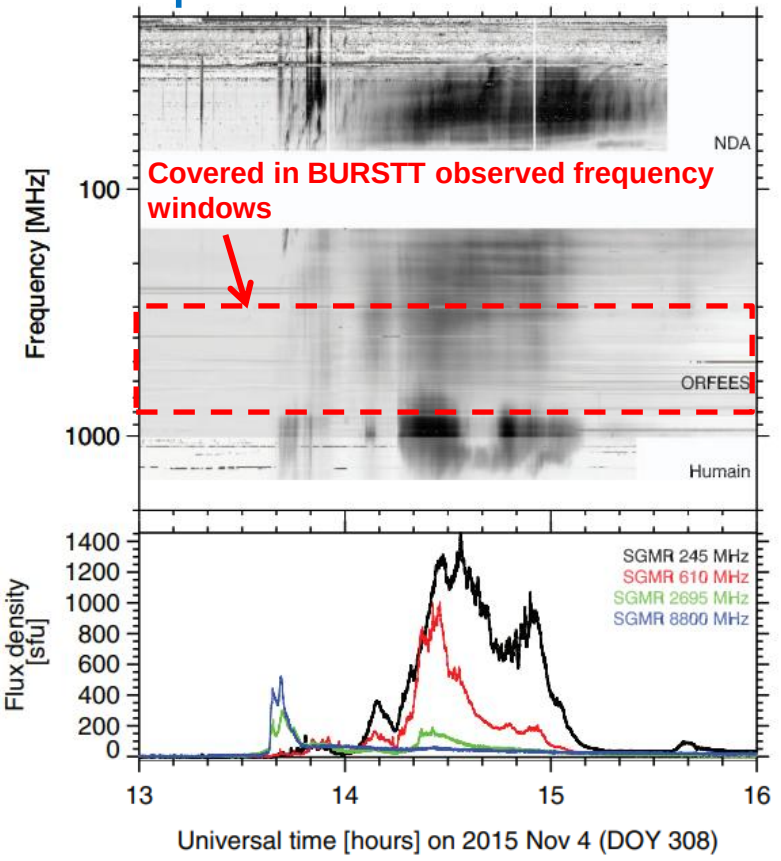


False echoes on ATC radar screen





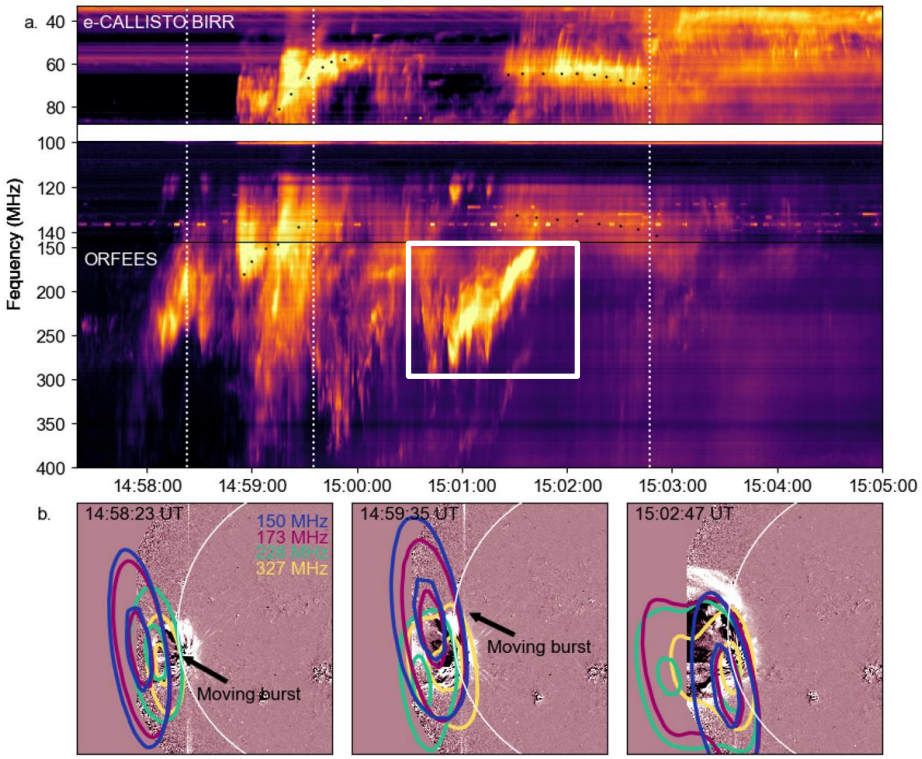


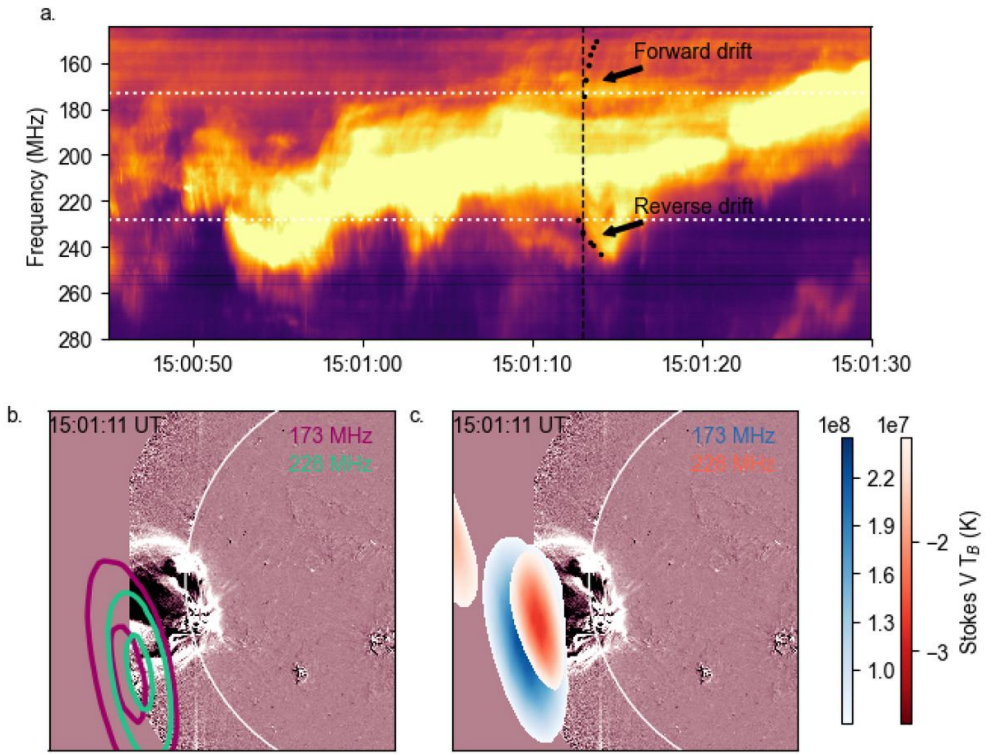


Morosan et al. (2022)

Imaging Spectroscopy of SRB

- Through imaging spectroscopy of SRBs, we can reconstruct the trajectories of energetic particles to better understand eruptive solar events, such as solar flares and Coronal Mass Ejections (CMEs).

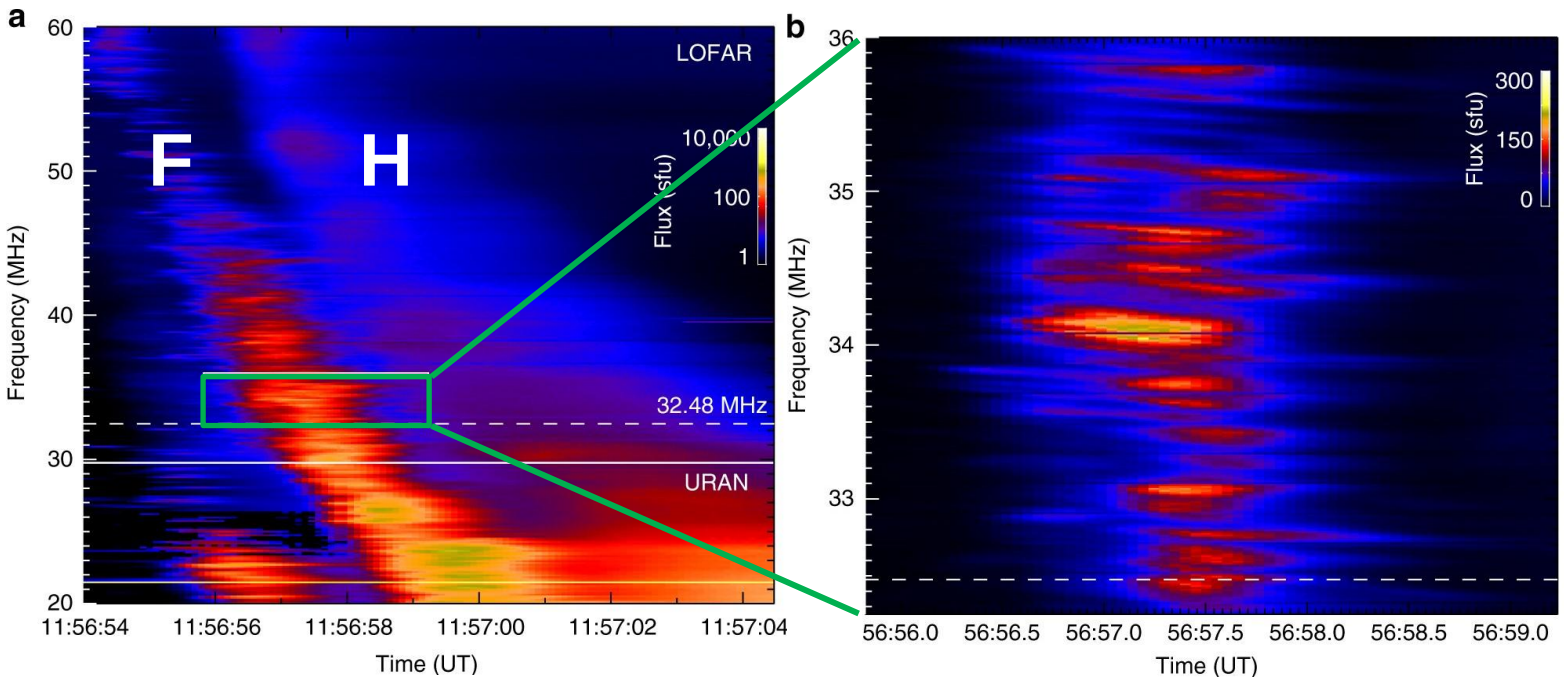


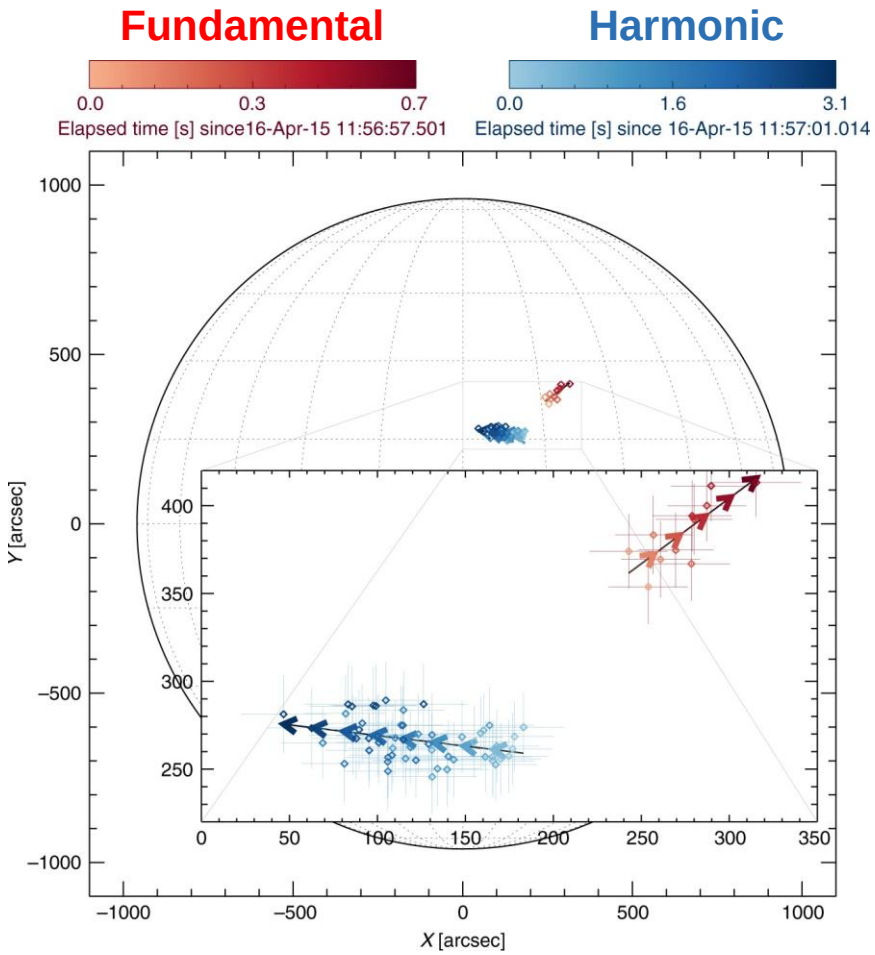


Kontar et al. (2017)

SRB Fine Structures

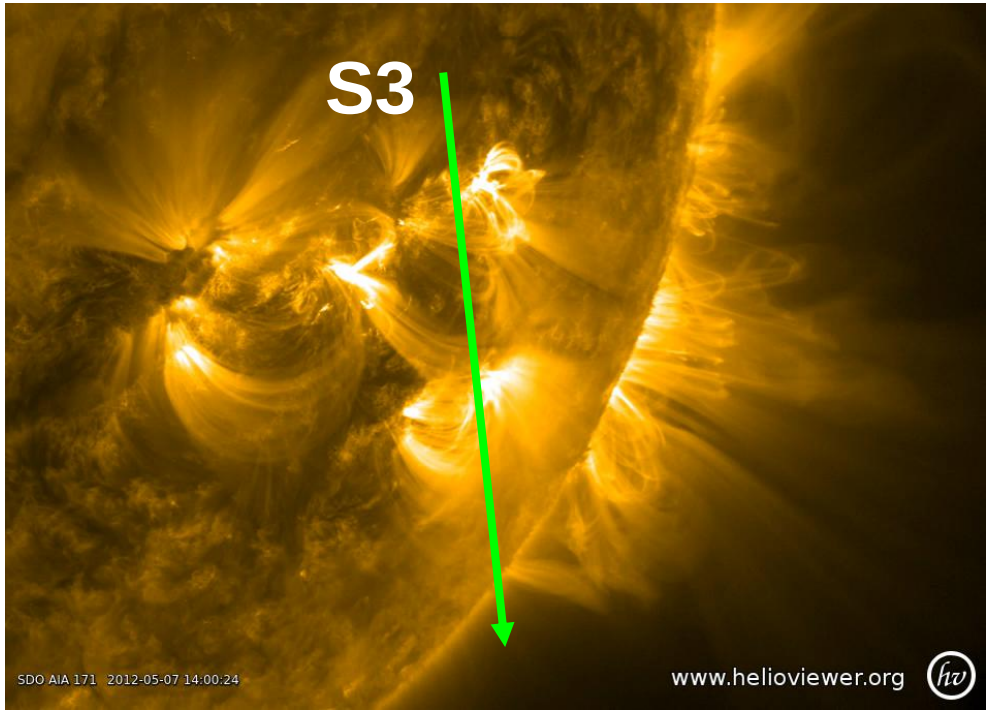
- Taking advantage of the high temporal resolution, we can distinguish between the F- and H- pairs of SRBs, and retrieve the propagation characteristics of energetic electrons for each component.

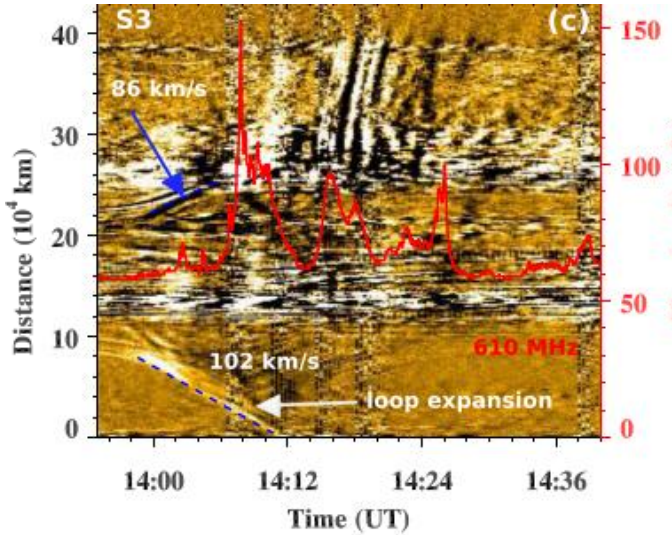


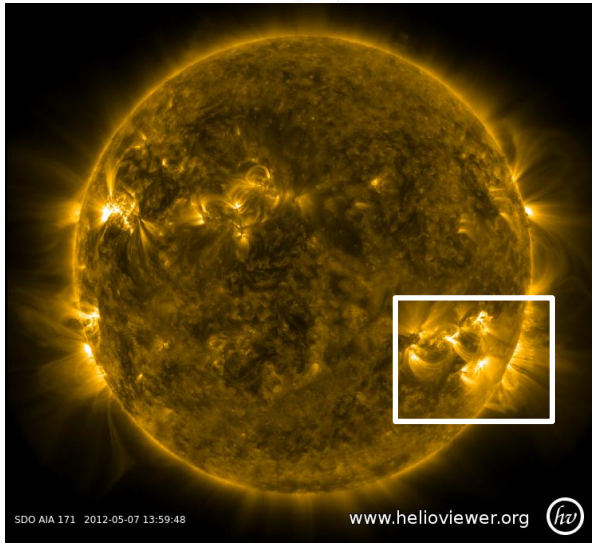


Quasi-periodic SRB Detection

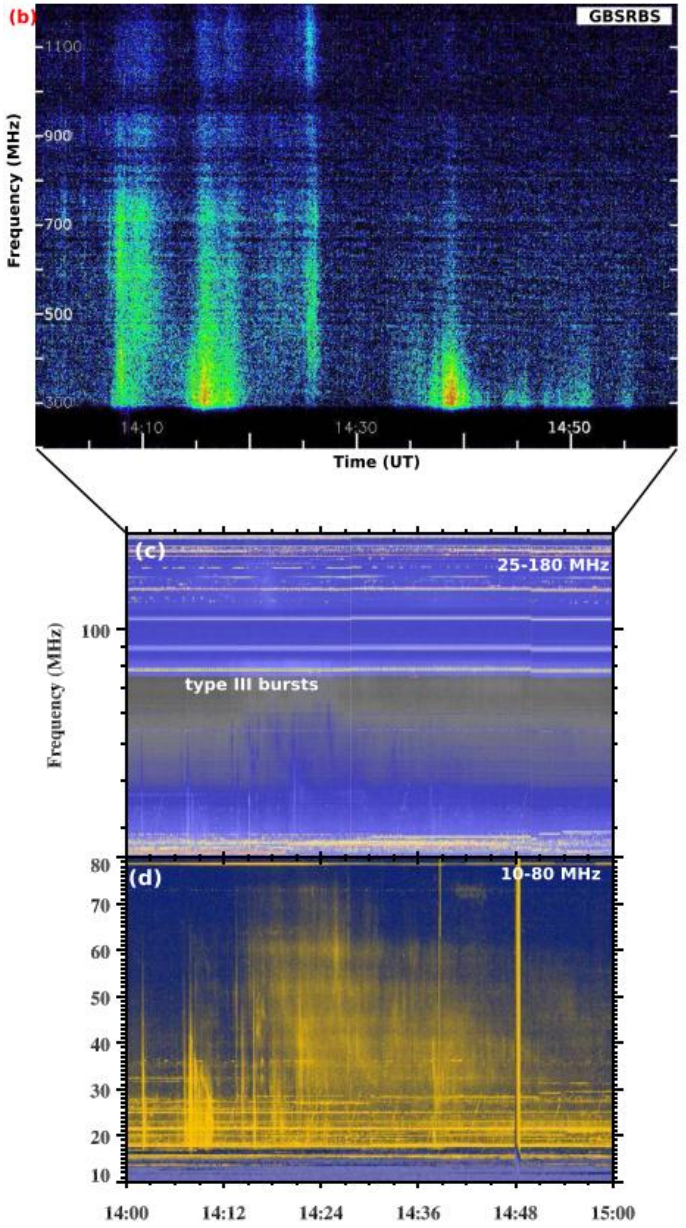
- Retrieved the possible magnetohydrodynamic (MHD) wave modes.







Kumar et al. (2017)

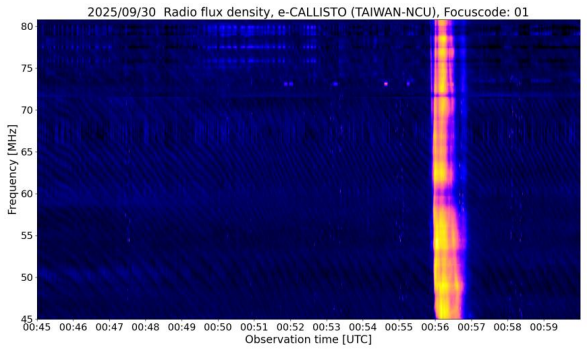


Current status of SRB research in Taiwan

Ground-base

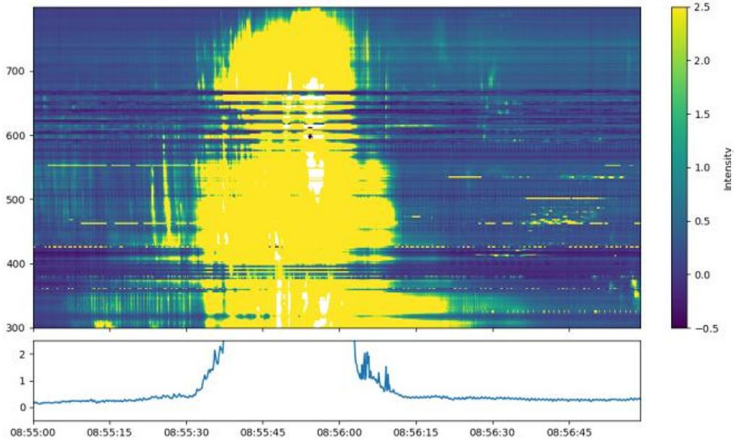
e-CALLISTO

45 MHz – 80 MHz



BURSTT

300 MHz – 800 MHz

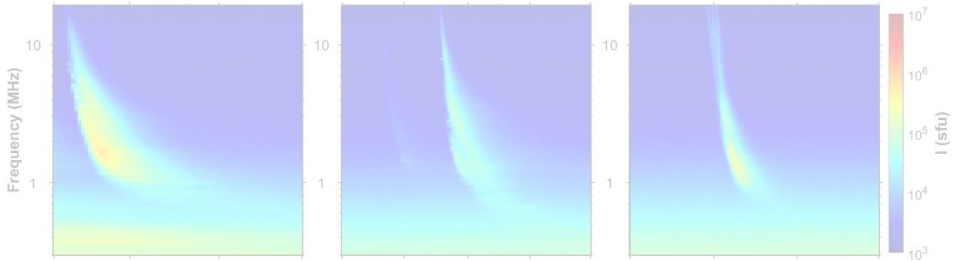


Spaceborne

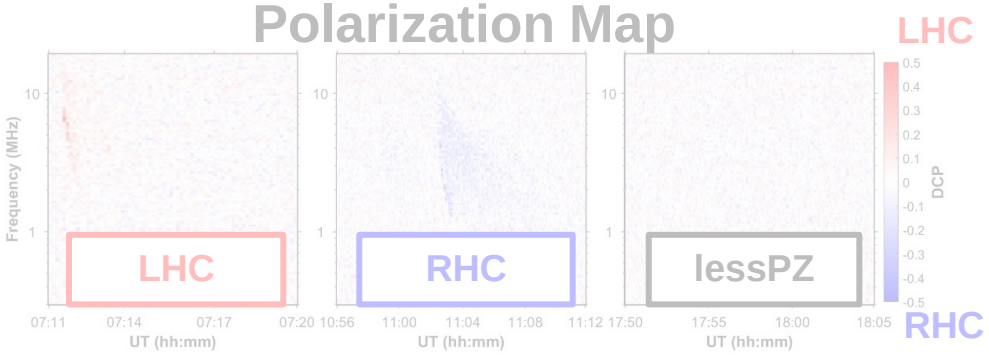
Parker Solar Probe

0.3 MHz – 20 MHz

Dynamic Spectrum



Polarization Map



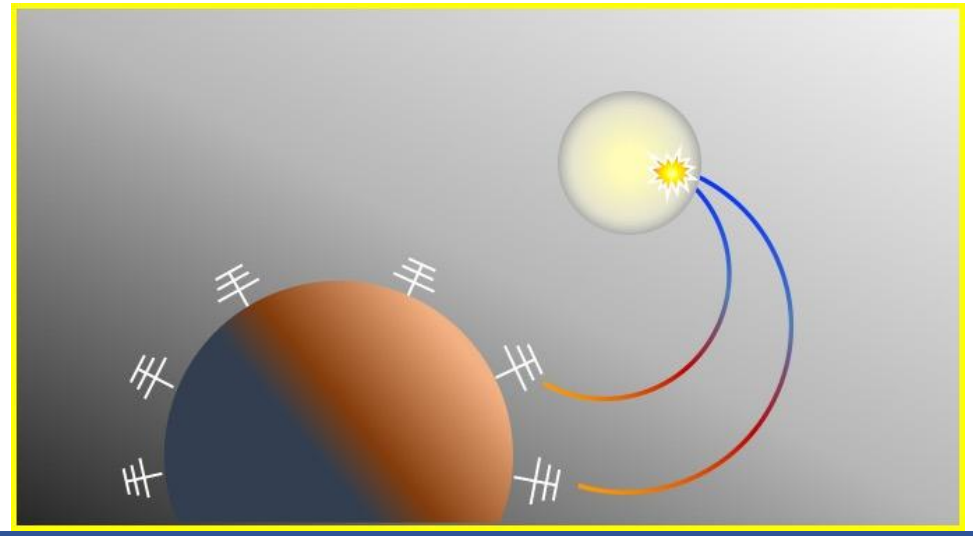
https://www.e-callisto.org/
https://astro-helio.ch/project/e-callisto/

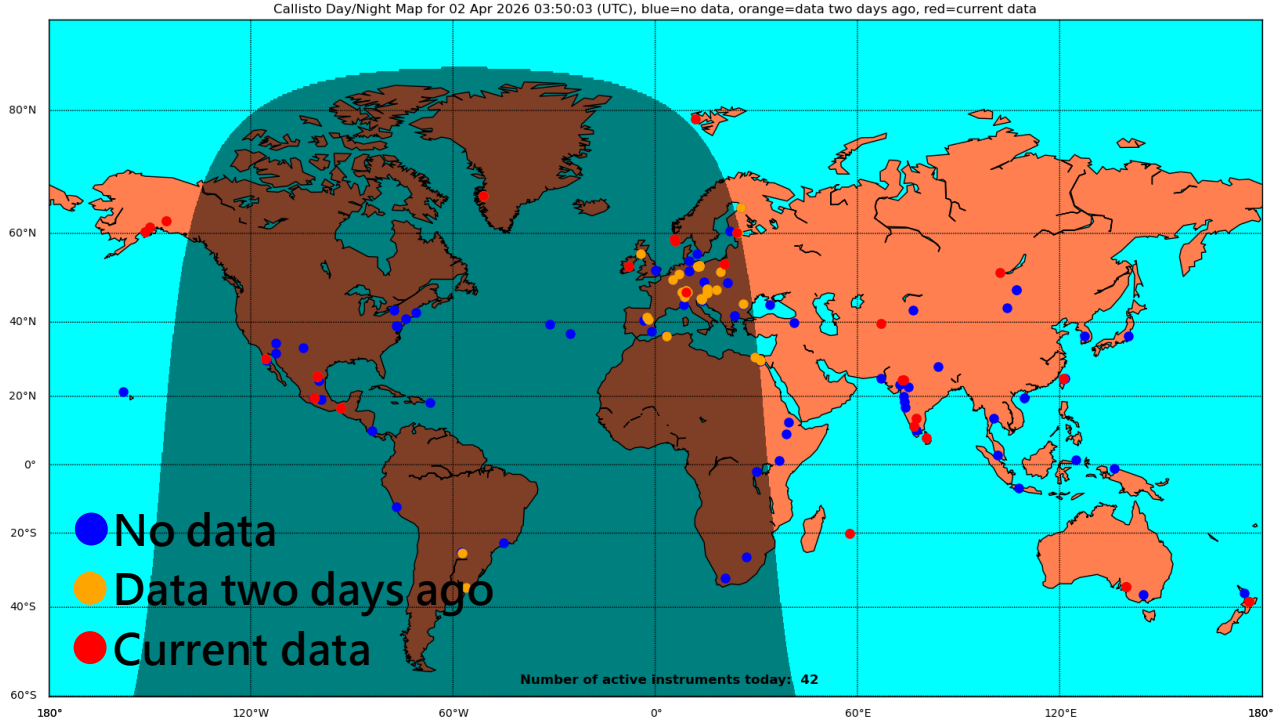
e-CALLISTO

Compound **A**stronomical **L**ow frequency **L**ow cost **I**nstrument for **S**pectroscopy and **T**ransportable **O**bservatory

International Network of Solar Radio Spectrometers, a Space Weather Instrument Array

- **Goal:** Understanding Transient Phenomenon in the Solar Corona
- **Frequency Range:** 45-870 MHz
- **Observed restrictions:** Only observed in the sunlit side
- **Number of Stations:** 236



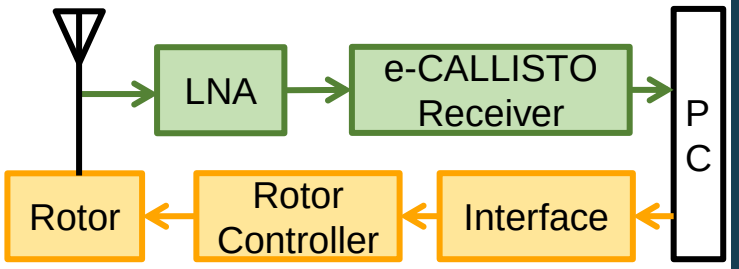


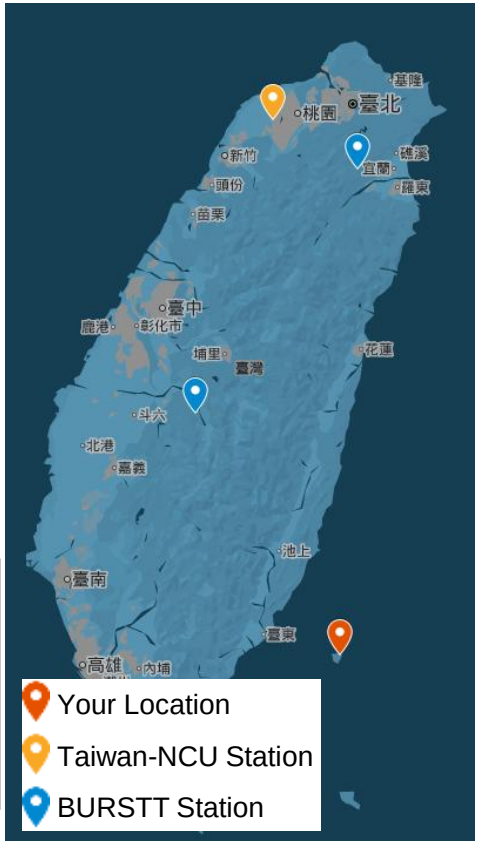
e-CALLISTO Taiwan-NCU Station

- Established since January 2022 at Chung-Li Very High Frequency Radar Station.
- The first and only one operational station in Taiwan on the e-CALLISTO network.

Geographical Coordinate	24.97º N, 121.19º E
Altitude	150 m
Time Zone	UTC+8
Daily Observation Duration	12 hr
Beamwidth of LPDA	60º-70º in the E-plane 110º-130º in the H-plane
Observed Frequency Range	45-80 MHz (since April 2024)
Number of Frequency Channel	200
Sweep Time	250 ms

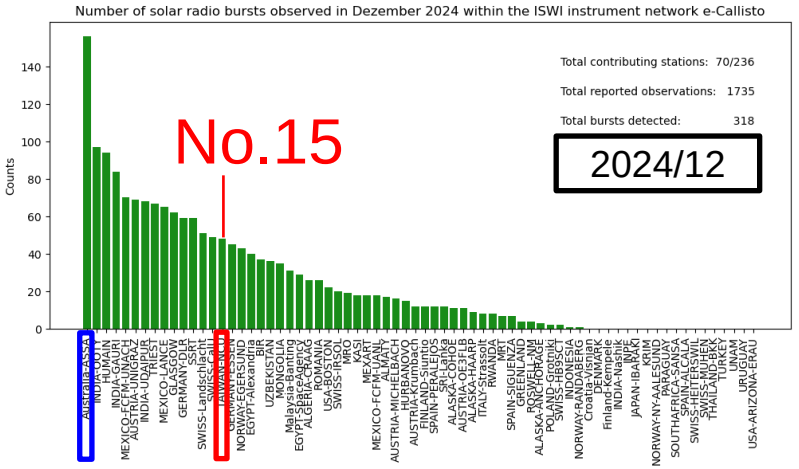


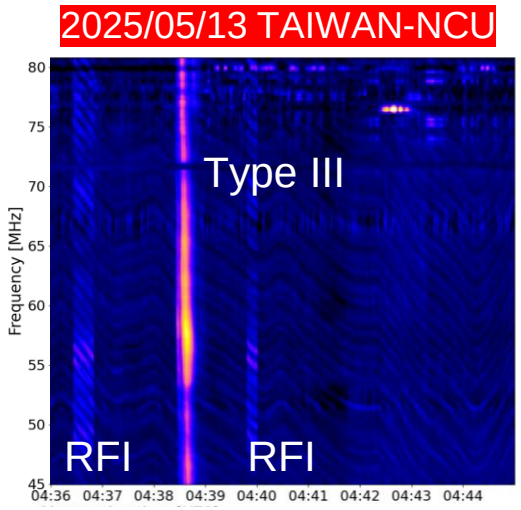


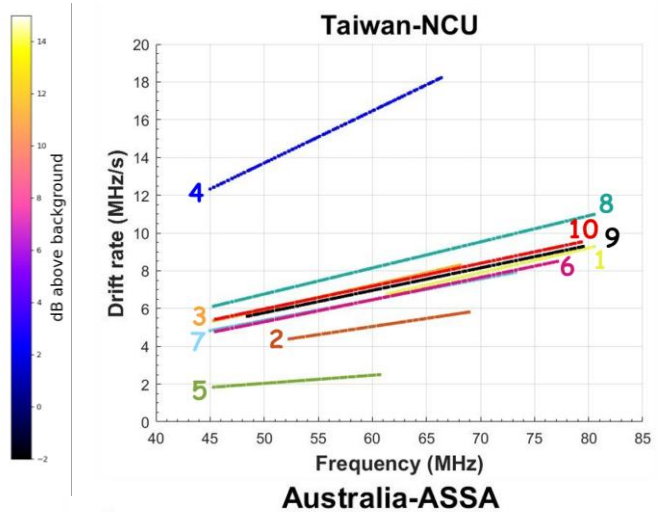


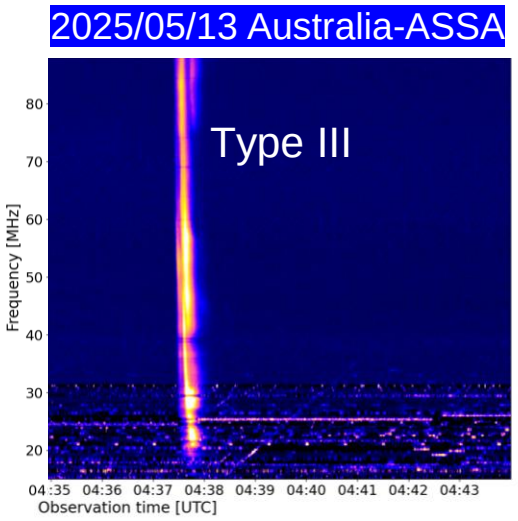
Drift Rate Estimation with SRB detection

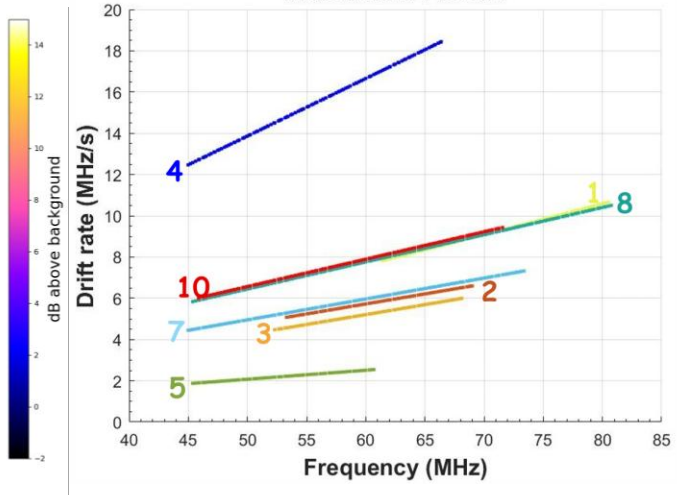
- Compared with the SRBs detected by the Australia-ASSA station, which recorded the highest number of SRB events in the e-CALLISTO network, the results indicate that events #4, 5, 7, and 8 exhibit similar drift rates and frequency dependencies between the Australia-ASSA and TAIWAN-NCU stations.





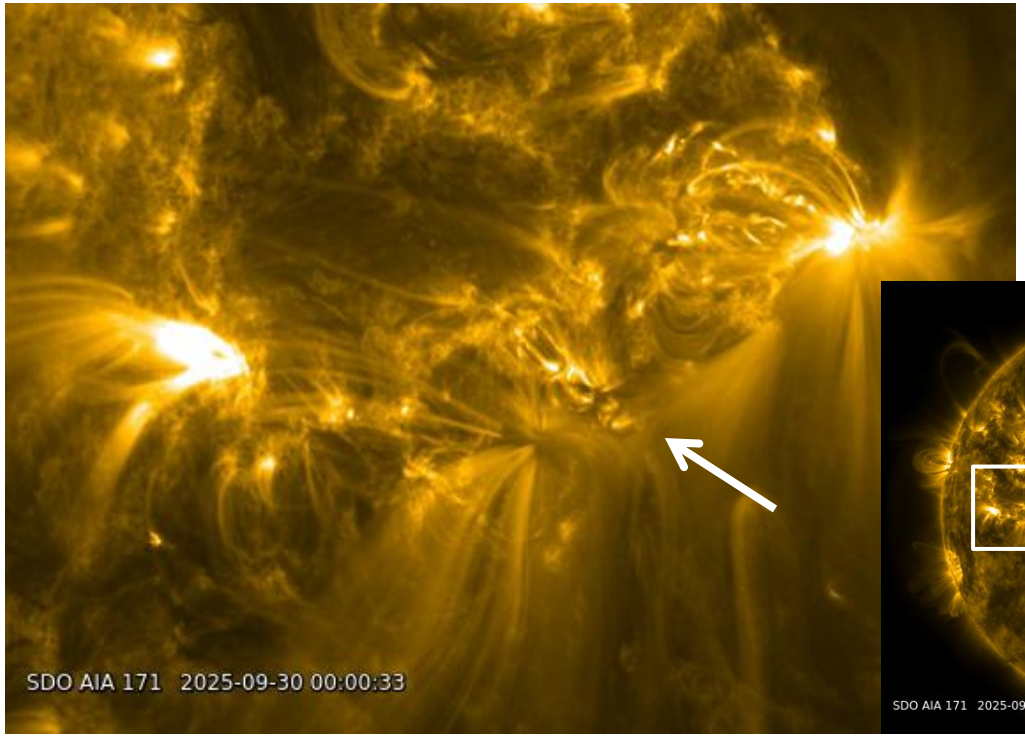


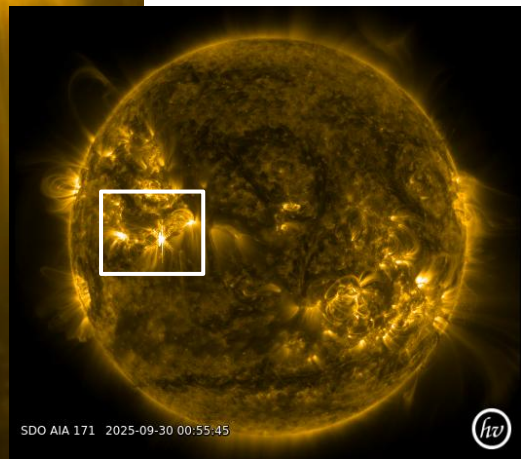


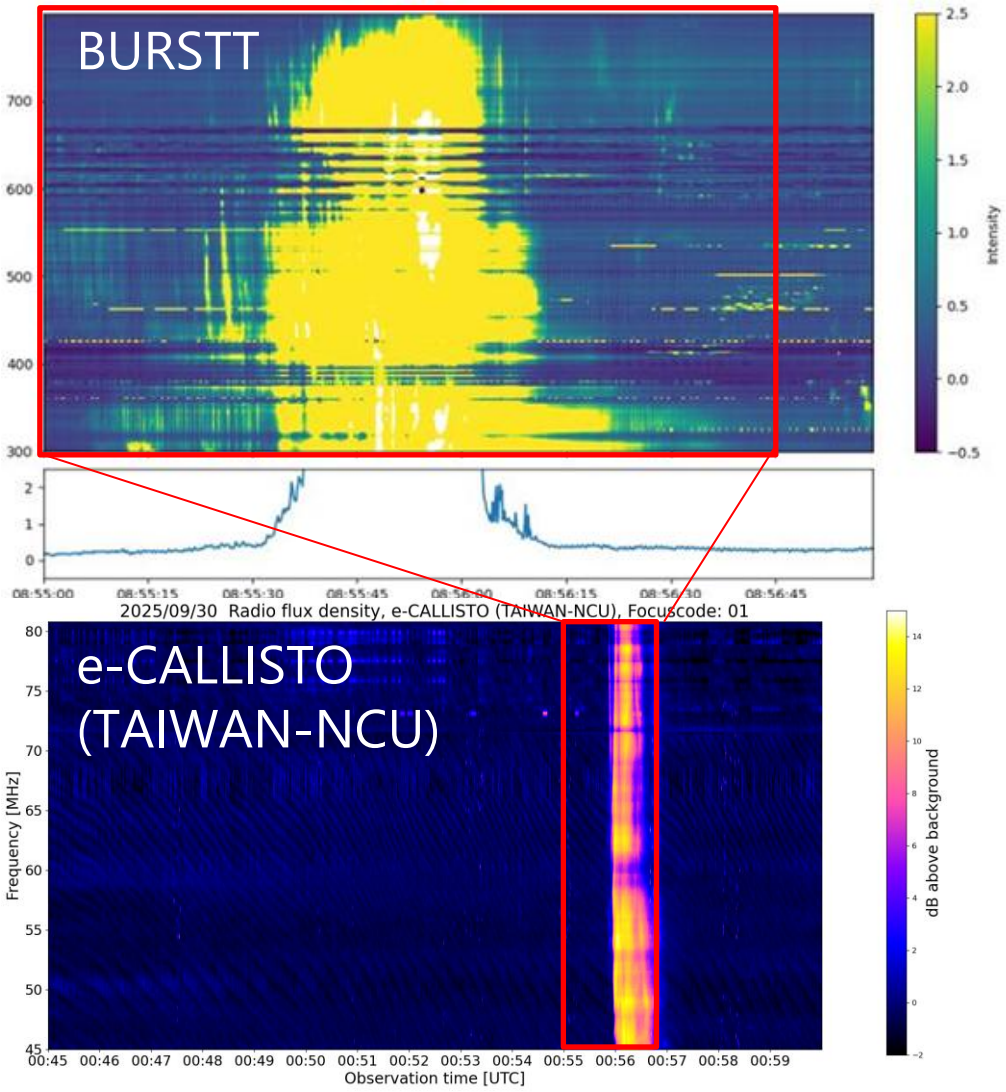


Solar Radio Burst Detected with BURSTT

- With high temporal capability, BURSTT could detected fine structure of solar radio burst.







Summary

- Similar to Fast Radio Bursts (FRBs), Solar Radio Bursts (SRBs) are short-lived, intense pulses of electromagnetic radiation in the radio frequency spectrum. However, unlike FRBs, the sources of SRBs originate exclusively from the Sun.
- SRBs are the most likely space weather candidates for explaining radar and GNSS interference. Therefore, it is necessary to better understand these sudden solar phenomena.
- With the high temporal and spatial resolutions of BURSTT, it is possible to reveal SRB fine structures, the propagation of energetic charged particles, and the mechanisms behind MHD wave modes.

9th Asia-Oceania Space Weather Alliance (AOSWA) Workshop in Taiwan!

- AOSWA encourages cooperation and shares information among institutes in Asia-Oceania region concerned with and interested in space weather.
- Dates: November 8 (Sun) – November 11 (Wed), 2026
- Venue: International Convention & Exhibition Centre Tainan (大台南會展中心, ICC Tainan)



Conveniently located in Southern Taiwan, within walking distance from the Tainan Station of Taiwan High Speed Rail (HSR).

See you at Nov 2026 in Tainan, Taiwan!

Tentative schedule

- 2026.04.13 Website open, Call for Participant
- 2026.06 Registration open
- 2026.08 Abstract deadline, review process
- 2026.09 Notification of acceptance/results

AOSWA 2026 Taiwan
Asia-Oceania Space Weather Alliance

Thanks for your listening!





https://swoo.cwa.gov.tw/

CENTRAL WEATHER ADMINISTRATION